

Skills

Languages: C++, C#, Python, C, GML, HTML/CSS

Technologies: Box2D, Arduino, Raspberry Pi, Git, Embedded Servers

Game Dev: Custom engines, Unity, GameMaker

Core Concepts: Developer Leadership, Custom Tooling, Image Processing Pipelines, Performance Optimization, Embedded Systems

Education

A.S. General Studies

- NCTC 2024

B.S. Computer Science

- UNT Expected 2027

Projects

Campaign Companion | *Founder & Lead Developer* | C#, Unity

- Founded **Hearthlight Studios LLC** to serve as an external client, securing 100% IP ownership while directing a capstone team on a full-scale TTRPG application.
- Led 2 concurrent student developer teams across Software Engineering and Image Processing courses with an additional 3rd team onboarding for capstone (10 total engineers); earned highest course evaluation and selected as model documentation for future classes.
- Built custom C# tooling in Unity, including an interactive map-to-note node graph system, a custom image processing biome brush pipeline, and a rich-text Obsidian markdown parser supporting bitmap clipboard ingestion.
- Growing an active beta testing community of 78 current members on Discord through regular devlogs and alpha releases.

2D Fantasy Platformer | *Team Project* | C++

- Built a complete level pipeline from authored data to runtime rendering and Box2D physics within a custom C++ engine.free
- Integrated LDtk level parsing and optimized rendering performance via camera-based tile culling.

Wireless DM-Controlled Magic Compass | *Embedded Systems Project* | C++, Arduino

- Developed C++ firmware and an embedded web server on constrained hardware for a wireless, servo-driven prop device.

Wireless DM-Controlled Datapad | *Embedded Systems Project* | Python, Raspberry Pi

- Designed handheld GUI software and a local web server with persistent storage on a Raspberry Pi for live gameplay management.

Portfolio

Scan to view Project Demos

